BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 318 · Paper 1 targeted reteaching

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does CIR hold?

Q2. What is a foreign key?

Q3. What is DNS used for?

Q4. Simplify $A \vee (\neg A \wedge B)$, showing working.

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I Do · Read new knowledge

Key vocabulary

reteach — Address an identified knowledge gap.

transfer — Apply a method to changed values.

method mark — Credit for correct intermediate reasoning.

New learning

Identify — Identify exact lost mark or incorrect idea

Model — Study fresh model with different values

Apply — Try an independent unseen example

Evaluate — Record the correction for exam revision

Worked example

A sequential fetch starts PC=120. Explain what MAR, MDR and CIR do.

1. Explain the method — MAR holds address, MDR transfers contents, and CIR holds instruction for decoding.

We Do · Practise with guidance

Show all steps for 8-bit two's complement −6.

`00000110 → invert11111001 → add1 11111010.`

Explain the method

Use exactly eight bits; inversion alone represents the wrong number.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

In a simple sequential fetch-decode-execute cycle, PC initially holds 70 and memory location 70 contains the instruction LOAD 25. State MAR, MDR, CIR and the next PC value, and explain one action of the control unit. Assume no branch or interrupt. [6]

Question 2

Convert −18 to eight-bit two's complement and explain sign. [6]

Question 3

Student(ID,Name) contains (1,Ada),(2,Ben). Mark(ID,Score) contains (2,80). Write an INNER JOIN using Student.ID=Mark.ID to select Name and Score, and state the output row. [6]

Exit Ticket · Answer before checking

Copying only the correct result proves a misconception has been resolved.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Current instruction.

Q2. Reference to a key in another table.

Q3. Resolving a domain name to an IP address.

Q4. $(A \vee \neg A) \wedge (A \vee B) = A \vee B$.

You Do pages · Independent questions

Q1. MAR=70 [1]; the instruction LOAD 25 is fetched from M[70] into MDR [2]; CIR receives LOAD 25 from MDR [1]; PC advances to 71 [1]; the control unit decodes the instruction in CIR [1].

Q2. +18=00010010 [2], invert11101101 [2], add1=11101110 [1], leading1 indicates negative [1].

Q3. SELECT Student.Name,Mark.Score [1] FROM Student [1] INNER JOIN Mark [1] ON Student.ID=Mark.ID [2]; output Ben80 [1].

Exit Ticket · Model answer

False. Explain the method on new values.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding